

Watch the effects of optimizing the hierarchy of the animatable object in the dialog [Show Graphic Structure \[button\]](#).

Note

As you cannot restore discarded data, only flatten the structure **after** you are done with modeling the object! You will not detect any difference in the appearance of the object; you will see a difference though in its graphic structure, which *Plant Simulation* flattened for better performance at runtime.

Only accept the result, when the optimized graphic meets your intentions in its looks and in its structure.

Back to [Visualizing the Material Flow](#)

Optimizing the Model

After we created our model and are sure that we do not have to make any more changes, we can optimize it.

Note

As you cannot restore discarded data, we advise to optimize the model only after you finished creating it. Optimizing the model improves performance.

We also recommend to save the model under a different name before optimizing it. This way you can return to the state the model had before optimizing it in case the optimized model reports errors or does not meet your expectations.

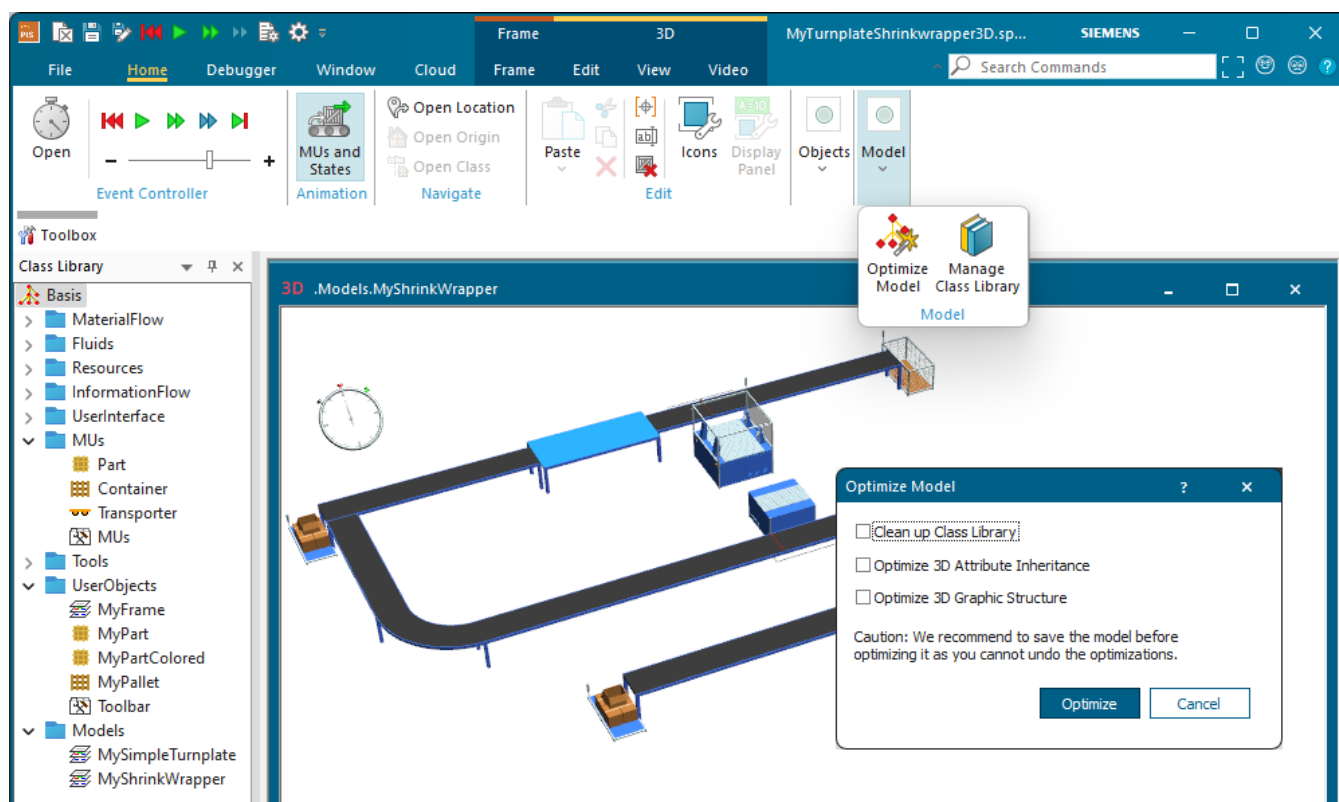
Proceed as follows:

- Save the model with a new name before optimizing it. This way you can return to the state the model had before optimizing it in case the optimized model reports errors or does not meet your expectations.



- Click **Optimize Model** on the **Home** ribbon tab.
- Select which optimization you want to execute. You can either choose only a single setting or any combination of the provided settings.

Below we demonstrate the effects of the individual settings with our model **MyTurnplateShrinkwrapper3D.spp**.

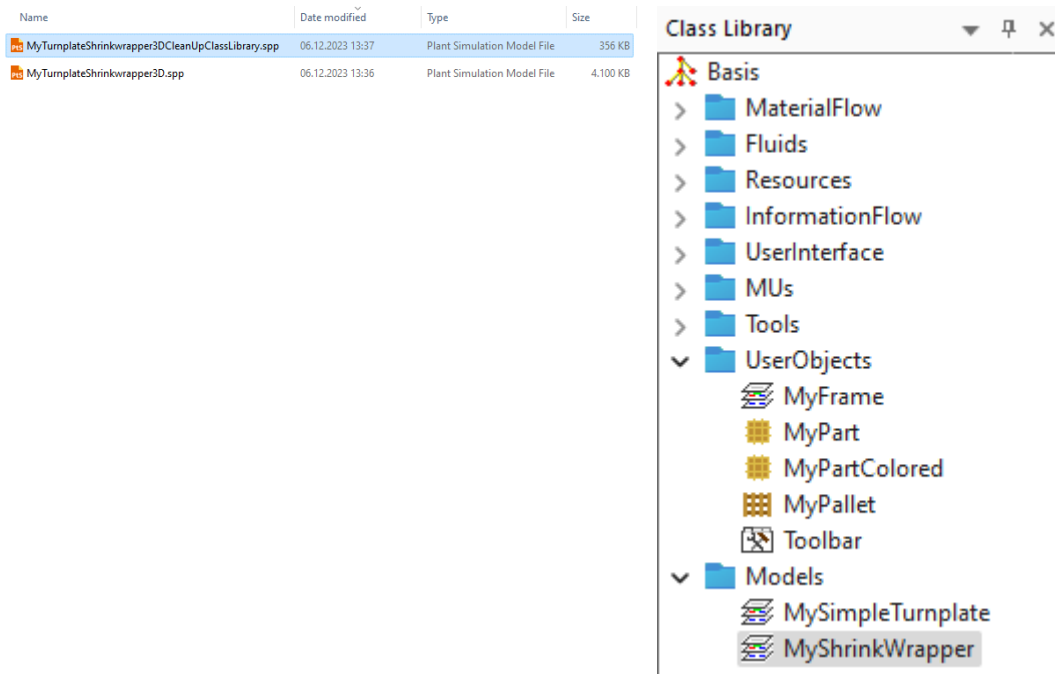


The finished model originally has this size.

Name	Date modified	Type	Size
 MyTurnplateShrinkwrapper3D.spp	06.12.2023 13:29	Plant Simulation Model File	4.108 KB

- First, we only select the setting **Clean Up Class Library**. This removes object classes from the *Class Library* that are not used in the model.

This setting reduces the file size the most.



If you want to continue working with the optimized model at a later point in time, you can re-add the missing objects with **Start > Manage Class Library**.

- Then, we only select the setting **Optimize 3D Attribute Inheritance**.

This setting reduces the file size in our example only marginally.

Name	Date modified	Type	Size
MyTurnplateShrinkwrapper3DAttributeInheritance.spp	06.12.2023 13:39	Plant Simulation Model File	4.100 KB
MyTurnplateShrinkwrapper3D.spp	06.12.2023 13:39	Plant Simulation Model File	4.100 KB

- Then, we only select the setting **Optimize 3D Graphic Structure**.

This setting reduces the file size in our example only marginally.

If you want to work with your model later on and would like to be able to edit your graphics, we recommend to not activate **Optimize 3D Graphic Structure**.

Name	Date modified	Type	Size
MyTurnplateShrinkwrapper3DGraphicStructure.spp	06.12.2023 13:42	Plant Simulation Model File	4.024 KB
MyTurnplateShrinkwrapper3DAttributeInheritance.spp	06.12.2023 13:41	Plant Simulation Model File	4.100 KB
MyTurnplateShrinkwrapper3D.spp	06.12.2023 13:39	Plant Simulation Model File	4.100 KB

- Finally, we combine all settings.

This results in the smallest file size which, alas, is only marginally smaller than the setting **Clean up Class Library**.

Name	Date modified	Type	Size
 MyTurnplateShrinkwrapper3DAIIOptimized.spp	06.12.2023 13:43	Plant Simulation Model File	332 KB
 MyTurnplateShrinkwrapper3DGraphicStructure.spp	06.12.2023 13:43	Plant Simulation Model File	4.024 KB
 MyTurnplateShrinkwrapper3DAttributeInheritance.spp	06.12.2023 13:41	Plant Simulation Model File	4.100 KB
 MyTurnplateShrinkwrapper3DCleanUpClassLibrary.spp	06.12.2023 13:38	Plant Simulation Model File	356 KB

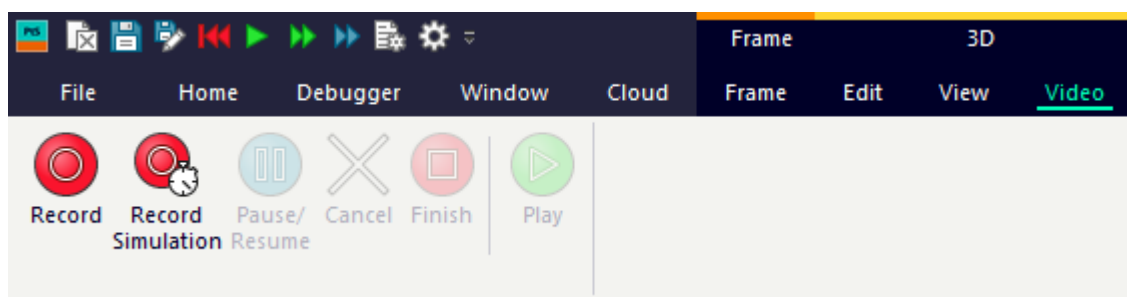
[Back to Visualizing the Material Flow](#)

Recording a Video

Record a Video

You can record a video of the crucial parts of your simulation model. This way, you can pass the video on to anybody or publish it on the Intranet or the Internet, without the person viewing the video having to install *Plant Simulation* on his computer. The recorder creates an AVI file that any multimedia player can play.

The [Video Ribbon Tab](#) provides all tools for recording and playing the video.

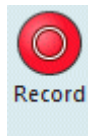
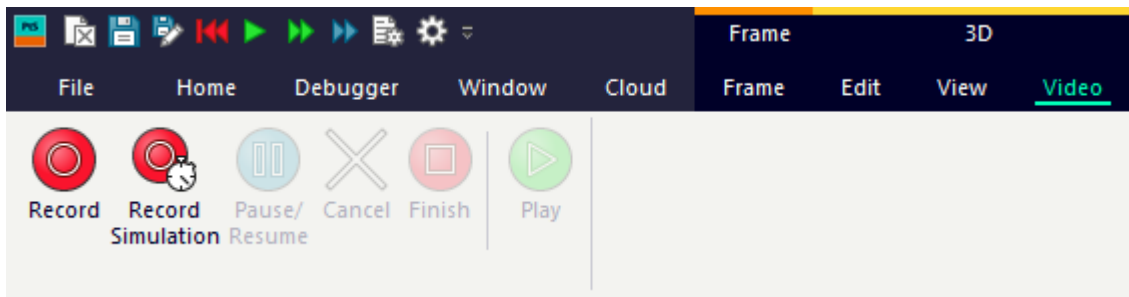


[Back to Visualizing the Material Flow](#)

Select Video Settings

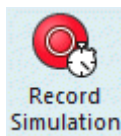
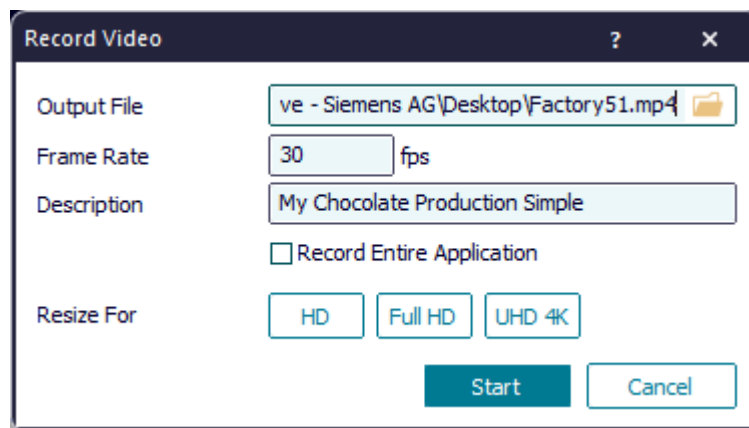
After you prepared the scene, you can select the video settings for recording the scene.

Proceed as follows:



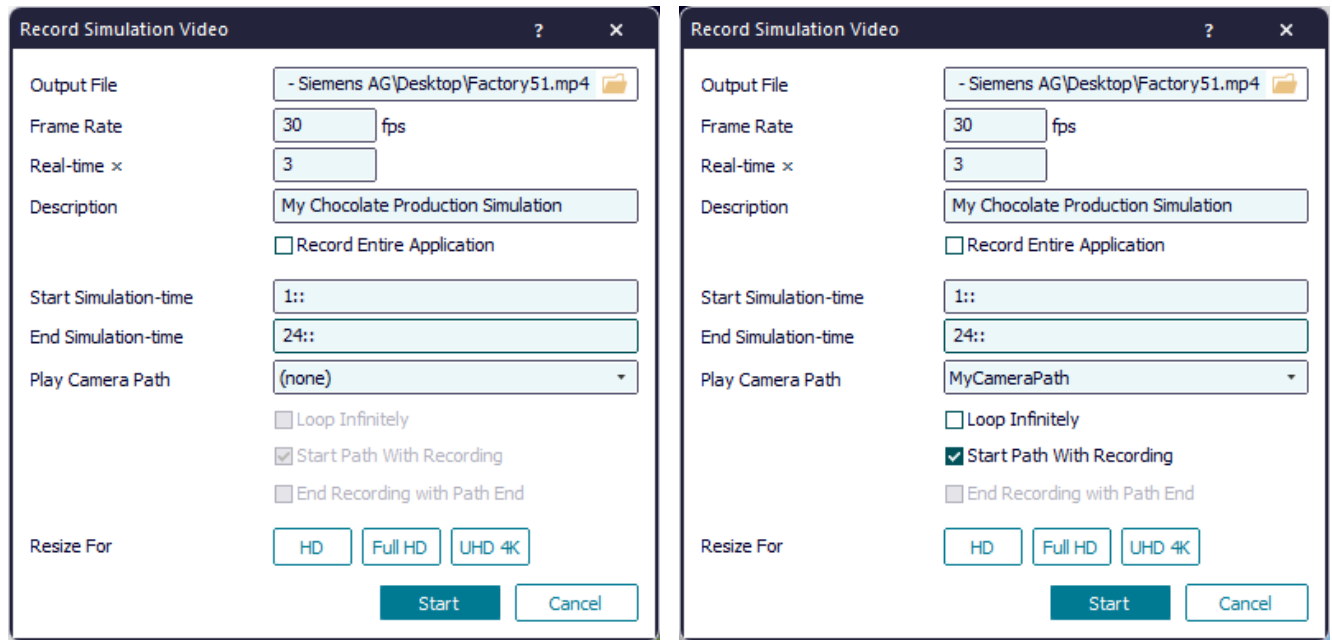
- Click **Start Simple Recording** on the ribbon tab **Video** and select the settings to record a video of *Plant Simulation* or the active 3D window.

The video will be recorded in real-time, meaning that recording a minute of physical real time corresponds to a minute of playback time in the video. You might, for example, use this type of video recording to demonstrate how to operate a model.



- Click **Start Simulation Recording** on the ribbon tab **Video** and select the settings to record a video of a simulation. The simulation does not have to run beforehand.

The video records in scaled simulation time, meaning that the number of minutes of simulation time, which you entered as **Real-time x [video]**, corresponds to one minute of playback time of the video. You can, for example, use this type of video recording to play the simulated production with realistic time behavior.

**Note**

To prevent that *Plant Simulation* overwrites the last recorded AVI file, rename the output file before recording another video.

Back to [Record a Video](#)

Set Up the Scene for Recording

There are a number of items to keep in mind **before** you start recording the scene.

Proceed as follows:

- Drag the window to a size that shows all that you want to show to the viewer, but not more, unnecessary information.

Note

If you would like to record a video in the display resolutions 720p or 1080p, you can set the size of the window to be recorded to these sizes in the dialogs [Start Simple Recording](#) and [Start Simulation Recording](#).

The size of the scene determines the size of the video, a large scene uses more processor time and might take a long time to animate and creates a larger video file.

- Decide from which position and which angle you want to view and record the scene. Then, rotate and zoom the scene accordingly before starting the simulation.

Note

If you are recording a simulation video, you can automatically connect the recording with playing a camera animation by selecting [Play Camera Path](#).

Back to [Record a Video](#)

Record the Video

After you set up the scene and selected the video settings, you can record the video by using the features on the **Video** ribbon tab.

- If you do not want to record the scene from the beginning of the simulation, start the simulation and let the animation proceed to about the time you want to start recording. Then, click  or  respectively to start recording.
- If necessary, reset the simulation model by clicking .
- To start the simulation, click **Start** .
- To start recording the animation, click  on the **Video** ribbon tab.

Note

Be aware that the recorder records the contents of the currently active window. If you click into another window during the recording session, or roll the mouse over a button to display a tooltip, these will show up in your recorded video as well. This might ruin the effect of your recorded video.

- To cancel recording, click .
 - To pause recording, click . This deactivates the **Pause** button.
You might pause recording if you do not want to show a certain section of the animation, and start recording the next section of interest by clicking  or  respectively again.
 - To stop recording the video, click .
- 3D show a dialog telling you that it saved the AVI file to the location and with the name you entered. Click **OK** to close the dialog and to continue working.

Note

If you want to view the video after you recorded it, we recommend to stop the simulation, as it uses a considerable amount of your computer's resources.

[Back to Record a Video](#)

Play the Video

After you have recorded the video, you can play it the way you recorded it. Or you can edit it with a digital video application, such as Adobe Premiere, Camtasia, etc.

Proceed as follows:

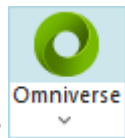
- To play the video, click  on the **Video** ribbon tab.
- The default player you set in MS Windows then plays the video.

[Back to Record a Video](#)

Visualizing the Simulation Model in Omniverse

Visualize the Simulation Model in Omniverse

You can visualize your *Plant Simulation 2606* simulation model in NVIDIA Omniverse using the *Omniverse Connector*.



The **Omniverse Connector** is only active if you have a subscription for this optional product.

To create an excellent looking visualization of your *Plant Simulation 2606* simulation model in Omniverse, proceed as follows:

- [Start the Omniverse Connector](#)
- [Set Up a New Filesystem Connection](#)
- [Establish a Live Connection to Omniverse](#)

You can then open the project in the Live data area of the *Nvidia Omniverse Composer* and work on enhancing the looks of the model.

- [Use Omniverse with the Function MVA_WriteUSD](#)

See also

[Omniverse Connector](#)

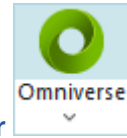
Videos

Start the Omniverse Connector

Start the **Omniverse Connector** to establish a connection with NVIDIA Omniverse.

Proceed as follows:

- Start *Plant Simulation 2606* and open the *Frame* containing the simulation model, which you want to use for exporting **usd** files or for a live connection with Omniverse.



- Change to the **Edit** ribbon tab and click **Omniverse Connector**.
- This opens the **LiveConnect** window.

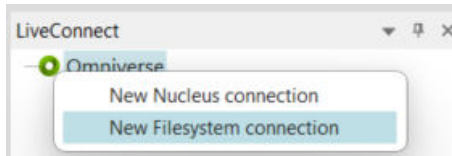
Back to [Visualize the Simulation Model in Omniverse](#)

Set Up a New Filesystem Connection

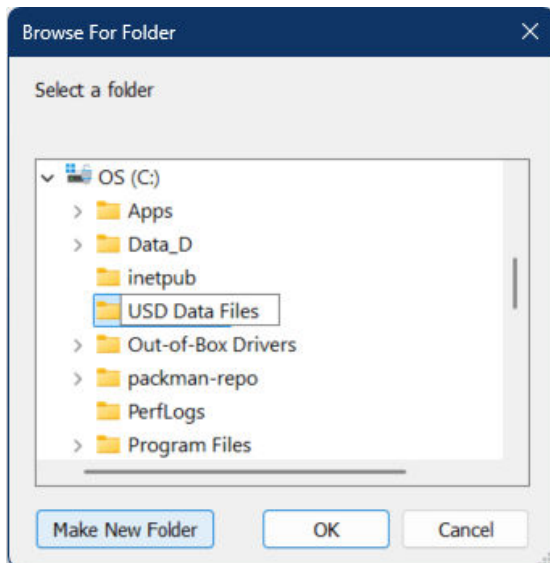
Set up a **New Filesystem connection** to export **usd** files and then establish a connection with Omniverse.

Proceed as follows:

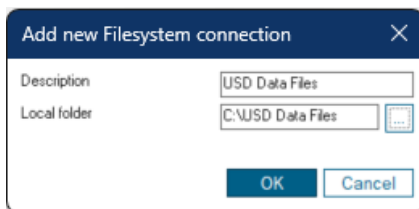
- Click the **Omniverse** node in the *LiveConnect* window and select **New Filesystem connection**.



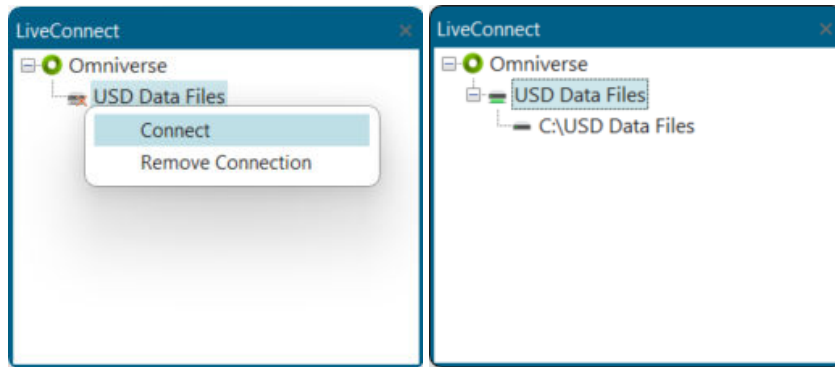
- Create a new folder in the file system on your computer by clicking the **Make New Folder** button or select an existing folder in the *File Browser*. Our folder is named **USD Data Files**.



- Type a **Description** of the project into the dialog **Add new Filesystem connection**.

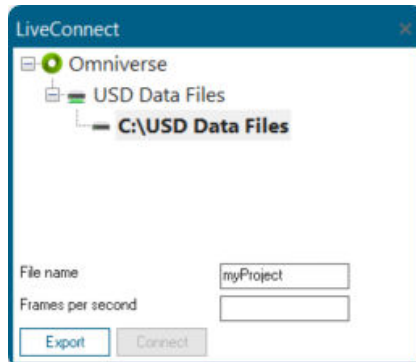


The **LiveConnect** window then shows our connection node.



- Click the **USD Data Files** folder and select **Connect** to establish the **Omniverse** connection with this folder. Then, select the data folder link. Once the folder is selected, you can type a file name into the text box **File name** at the bottom of the **LiveConnect** window.

Plant Simulation does currently not use the setting **Frames per second**, you can leave this box empty.



Export the Simulation in USD Format

After setting-up the file system connection, you can export the running simulation to **usd** files. **USD** stands for **Universal Scene Description**.

Proceed as follows:

- Click **Export** to make Omniverse create the required **usd** files within the configured data folder.
Start the simulation, which you want to record in *Plant Simulation*.
Stop the simulation after everything is recorded that you want to show.
- Click **Export** again to stop the **usd** recording,
- Plant Simulation* creates a library folder and a **usd** file in the data folder.
- You can then open the exported simulation project in the *Nvidia Omniverse Composer* and visualize the simulation model.

Back to [Visualize the Simulation Model in Omniverse](#)

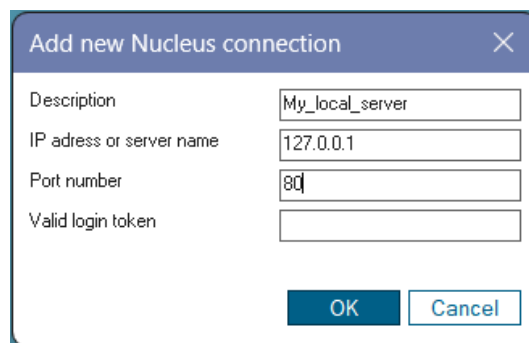
Establish a Live Connection to Omniverse

Establishes a live connection to **Omniverse** using the **Omniverse Nucleus Server**.

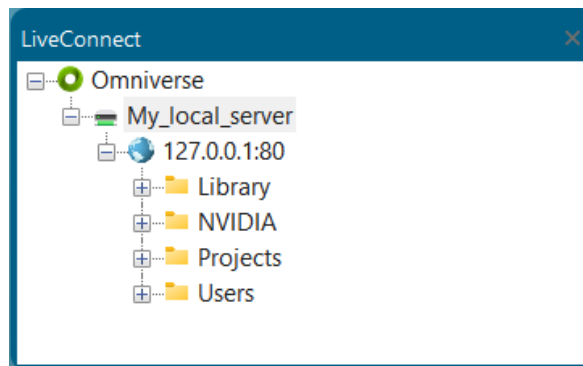
Proceed as follows:

- Make sure that the **Omniverse Nucleus Server** is started and running.
- Right-click the Omniverse node in the **LiveConnect** window and select **New Nucleus connection**.
- Type in a **Description** and the **IP address or the server name** of the computer on which Omniverse Nucleus is installed.

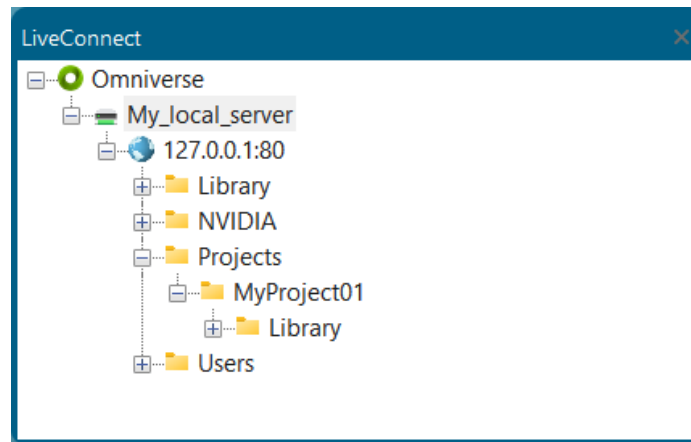
Type in the **Port number**, which is configured for the Omniverse Nucleus Server. Click **OK** to close the dialog.



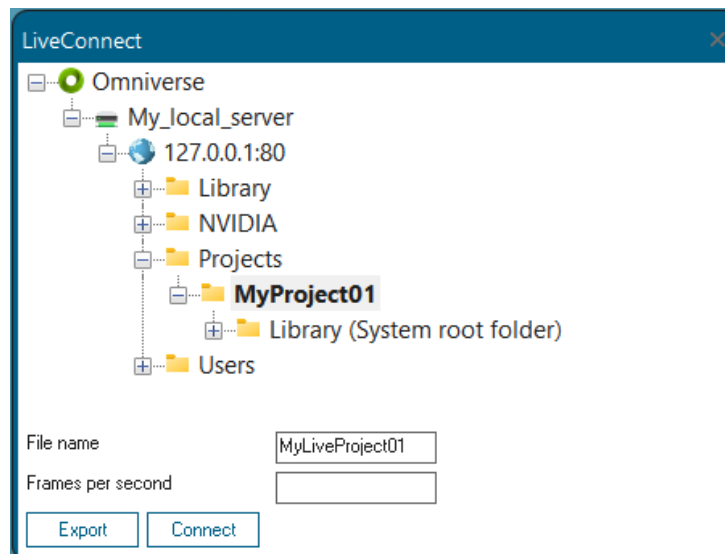
- Right-click the newly created server node and select **Connect** on the context menu.
- You will then see the Omniverse data folder on the server computer.



- Create a folder structure in *Nvidia Omniverse Composer* in the **Projects** folder .



- Right-click the **Library** folder in the project, **MyProject01** in our example. Then select **Set System root folder** on the context menu.
- **Select the project folder** and type a project name into the text box **File name**. We typed in **MyLiveProject01**.



- Click **Connect** and start the simulation in the *Frame* containing your simulation model.
- Open the project in the Live data area of the *Nvidia Omniverse Composer*.
- Stop the simulation in *Plant Simulation* and then terminate the Live connection by clicking **Connect** in the **LiveConnect** window.

Back to [Visualize the Simulation Model in Omniverse](#)

Video

https://support.sw.siemens.com/en-US/knowledge-base/KB000179496_EN_US

Use Omniverse with the Function MVA_WriteUSD

Creates and activates a *Nucleus Connection*, or creates and activates a *Filesystem Connection*, or terminates a *Live Connection*, or a *Filesystem Connection* via *SimTalk*.

Remarks

Before using *MVA_WriteUSD*, you

- Either have to load the Omniverse DLL via SimTalk with

```
loadLibrary(applicationHome + "PlantSimOmniverse.dll")
```

- Or you have to manually start the Omniverse connection by clicking **Omniverse Connector** on the **Edit** ribbon tab in *Plant Simulation*.



Type

Function

Syntax

```
MVA_WriteUSD("parameter string")
```

Create and Activate a Nucleus Connection

Specify these instructions to create and activate a *Nucleus Connection* with *MVA_WriteUSD*:

```
var system_root_path:string = "/Projects/My_Project01/Library"
var output_path:string = "omniverse://127.0.0.1:80/Projects/My_Project01/
myExport.live"
MVA_writeUSD(output_path+"?sysroot="+system_root_path)
```

Parameter

The parameter *string* specifies the **project file path** in the first section and the **Omniverse system root folder** in the second section, which is to be used for the USD export or a live connection.

```
MVA_writeUSD(<project file path>?systemroot=<system root path>)
```